

Credit Policy Recommendation Report

CAC 2026

Portfolio: 50,000 Customers
70,000 Loans (Jan 2021 – Jun 2024)

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1 Executive Summary

This report presents findings from a comprehensive quantitative analysis of our digital lending portfolio and introduces a machine-learning-powered Early Warning System (EWS) for 30-day forward delinquency prediction.

Our analysis reveals that the overall portfolio default rate stands at **10.8%**, with a Loss Given Default (LGD) averaging **28,647** and a mean recovery rate of **47.3%**. Approximately **68.2%** of loans are profitable, but the distribution masks a deeply unprofitable tail concentrated in the Subprime and Thin-File segments. The EWS model’s Amber+Red buckets capture **99.7%** of all loans that will go 30+ DPD in the next installment cycle, providing collections teams with a reliable, actionable signal. Recall has improved from 35% (v1.0) to 67.5% — nearly doubling the delinquency catch rate — while false negatives have been cut from 6,343 to 3,170.

2 Data Description

This study uses a synthetic digital lending dataset modelled on realistic Indian lending market conditions, spanning the period January 2021 to June 2024. The dataset comprises six interlinked tables covering the complete loan lifecycle — from customer onboarding to final repayment outcomes — totalling over 2.5 million records across all tables.

Table	Records	Description
Customers	50,000	Borrower demographic and credit profiles
Loans	70,000	Loan and product-level details
Repayments	1,144,062	Monthly EMI payment events
Behavioral Signals	1,236,752	Monthly bank transaction indicators
Outcomes	70,000	Loan-level financial performance
Master (Joined)	70,000	Wide analytical table (Loans + Customers + Outcomes)

Table 1: Dataset Overview

The dataset comprises 50,000 borrowers and 70,000 loans across 20 Indian cities spanning Metro (38%), Tier-2 (36%), and Tier-3 (26%) geographies. Borrowers are classified into four risk tiers: Prime (28%), Near-Prime (37%), Subprime (25%), and Thin-File (10%). The loan portfolio includes six product categories (Personal Loan, BNPL, SME Working Capital, Two-Wheeler Loan, Consumer Durable, and Education Loan) distributed through seven acquisition channels. Loan amounts range from Rs.1,000 to Rs.20 lakh, with approximately 28% of customers holding multiple loans.

The repayment dataset records 1.14 million monthly EMI events, including payment status, Days Past Due (DPD), and RBI-standard loan classifications. Payment deterioration is simulated to begin up to three months before default, enabling early-warning analysis. The behavioral signals dataset contains 1.24 million monthly observations capturing cash-flow consistency, balance volatility, income and spending shocks, average inflows, EMI obligation ratio (FOIR proxy), and active loan counts.

The outcomes dataset summarizes loan-level financial performance, with an overall default rate of 10.8%, recovery rates ranging from 25–65% for defaulted loans, and approximately 68.2% of loans generating positive customer lifetime value after accounting for acquisition, operating, and credit-loss costs.

3 Risk Segmentation Analysis

3.1 Portfolio Composition & Default Rates by Segment

The portfolio comprises four origination-time risk tiers. Default rates are monotonically ordered by tier — a strong validation of the origination scoring methodology — but the Subprime segment’s 20.2% default rate and the Thin-File segment’s 15.4% default rate represent disproportionate credit loss concentrations that demand targeted policy intervention.

Segment	N (Loans)	Share	Default Rate	Avg Income ()	Bureau Score	Profitable
Prime	18,809	26.9%	3.1%	1,03,691	729	78.2%
Near-Prime	24,919	35.6%	11.0%	51,217	649	74.2%
Subprime	16,034	22.9%	20.2%	26,896	559	64.2%
Thin-File (NTC)	6,599	9.4%	15.4%	22,100	0 (no bureau)	64.2%

3.2 Feature Predictive Power (Information Value Ranking)

IV (Information Value) is the gold standard for scorecard feature selection. Any feature with $IV > 0.3$ must be a core model input; anything < 0.02 must be excluded. The IV ranking directly tells you the priority order for engineering features: Risk Grade $\rightarrow$ Bureau Score $\rightarrow$ Employment Type $\rightarrow$ Income $\rightarrow$ APR. IV analysis confirms that Risk Grade, Bureau Score, and Employment Type are the strongest origination-time predictors. Monthly Income and APR contribute additional medium-strength signal. Geographic tier and acquisition channel, while statistically significant, carry lower standalone predictive power.

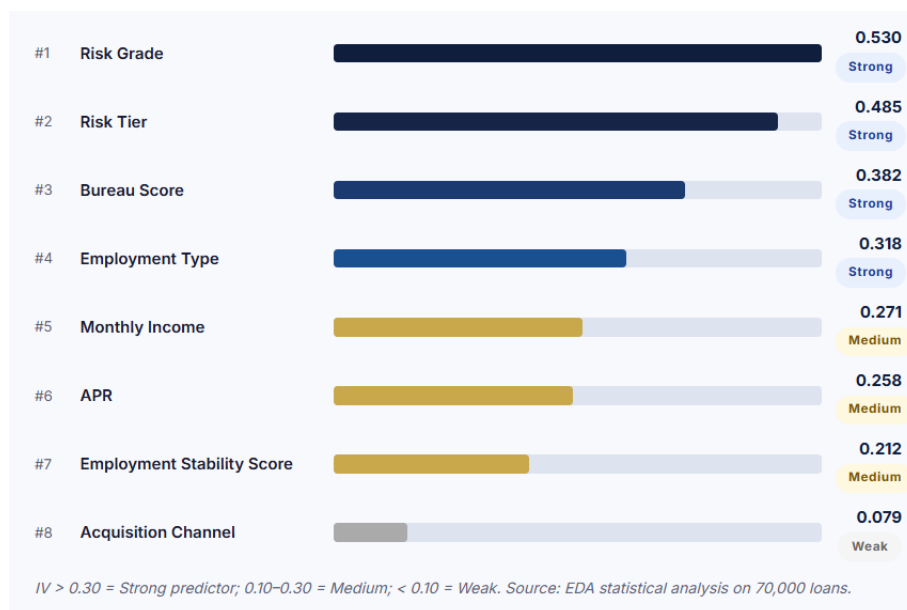


Figure 1: Feature Importance Value

3.3 Channel Risk & Acquisition Cost Dynamics

Acquisition channel analysis reveals a compounding risk: Paid-Digital and DSA-Agent channels simultaneously carry the highest Customer Acquisition Cost (CAC) and the highest default risk premium. These channels currently represent 48% of loan volume ($19,562 + 14,105 = 33,667$ loans) but are structurally diluting portfolio ROE. This creates a double drag on portfolio unit economics that is not offset by their volume contribution.

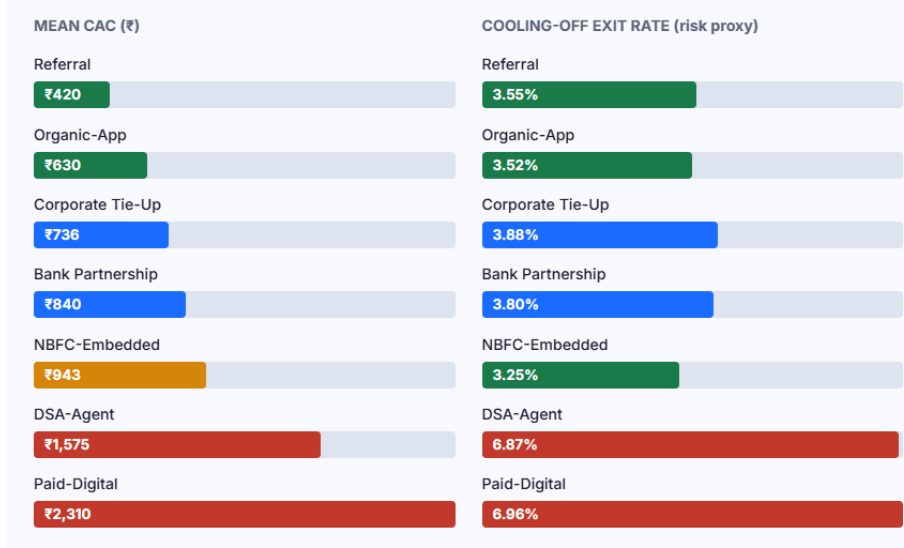


Figure 2: Feature Importance Value

3.4 Product Mix, Ticket Sizes & Tenure Strategy

3.4.1 Product-Level Profile

Product type is significantly associated with default (Chi-square $p < 0.001$, Cramer's $V = 0.035$), but the pattern is nuanced. BNPL has the lowest ticket size, so the absolute loss per default is limited. Personal Loan is the most critical product to manage because its large median ticket and high volume make it the largest contributor to portfolio NPA (Non-Performing Asset) exposure. SME-Working-Capital shows the highest skew, meaning a small number of large loans drive most of the risk. Two-Wheeler Loans are relatively safer because collateral reduces LGD and provides recovery support.

Product	Volume	Median Ticket (Rs.)	Risk Signal
BNPL	16,574	25,000	Moderate volume, low per-loan loss
Consumer-Durable	8,554	50,000	Low skew, predictable
Two-Wheeler Loan	8,034	82,000	Collateral-backed, lowest risk profile
SME-Working-Capital	9,047	1,15,000	High tail risk due to concentration of large loans
Personal Loan	24,313	1,33,000	Largest segment, high concentration risk
Education Loan	3,478	1,41,000	Long tenure, high LGD potential

Table 2: Product-level profile across volume, ticket size, and risk signal

3.4.2 Ticket Size – The Optimal Band

Overall portfolio ticket size is right-skewed, with a median of Rs.74,000 and a mean of Rs.1,05,047. Defaulted borrowers had 21% lower ticket size than non-defaulted borrowers, indicating that smaller loans at origination often reflect lower income or weaker creditworthiness rather than lower risk. Ticket-to-Income Ratio is more informative than ticket size alone.

Segment	Recommended Ticket Range	Policy Rule
Prime (bureau 700+)	Up to Rs.3,00,000	High income can absorb EMI obligation
Near-Prime (bureau 600–700)	Rs.50,000–Rs.1,50,000	Monitor FOIR < 40%
Subprime (bureau 500–600)	Cap at Rs.75,000	Enforce FOIR < 35%
Thin-File (NTC)	Rs.10,000–Rs.30,000	Start with BNPL / Consumer-Durable ladder

Table 3: Suggested ticket sizing by borrower segment

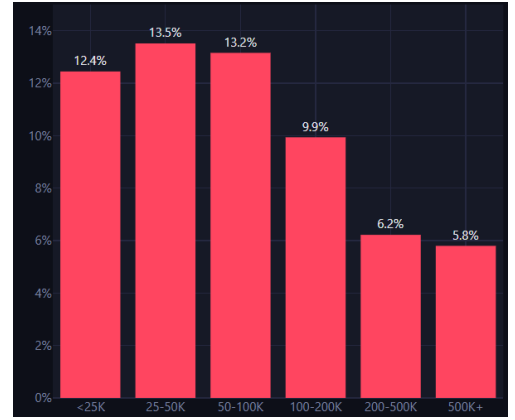


Figure 3: Default rate by ticket-size band

3.4.3 Tenure Strategy– Recommendation

The portfolio has an average tenure of **18.6 months**. Defaulted borrowers exhibit slightly shorter average tenures (17.8 months versus 18.7 months), suggesting that a significant proportion of defaults occur early in the repayment cycle. This indicates that **loan age is highly relevant for in-life risk monitoring and early warning systems**. Furthermore, short-tenure products such as BNPL provide limited time for behavioural signal collection, necessitating alternative scoring and monitoring approaches.

Segment	Recommended Tenure Cap	Rationale
BNPL / Consumer-Durable (Subprime)	12–18 months	Short exposure window limits LGD; quick assessment of repayment behavior
Personal Loan (Near-Prime)	24–36 months	Balanced EMI affordability versus portfolio vintage exposure
Personal Loan (Subprime)	Max 24 months	Restricting tenure limits interest income loss on default
SME-Working-Capital	Linked to cash-flow cycle (3–18 months)	Avoid long-tenure unsecured SME exposure
Education Loan	Up to 60 months	Moratorium periods justified; monitor post-study income

Table 4: Recommended tenure strategy by product segment

4 Early Warning System Findings

4.1 Signal Architecture

The EWS model is built on the empirically validated finding that financial stress manifests in bank behavioral signals 1–3 months before payment deterioration and 3–6 months before formal default. The model processes three layers of signals in a rolling 3-installment window as shown in the Fig 4:

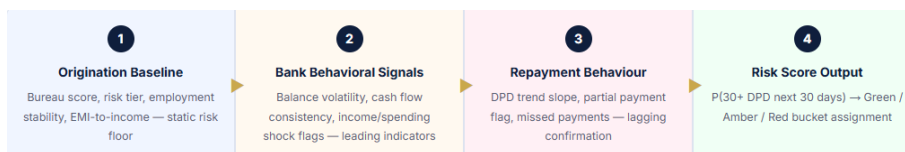


Figure 4: EWS model architecture

4.2 Pre-Default Behavioral Signal Deterioration

Our data shows clear deterioration across all three key behavioral signals in the months leading up to a missed payment — with stress building as early as 4–6 months before default and accelerating sharply in the final 3 months. This pattern confirms that the 3-month rolling window used by the EWS model captures the critical warning period, giving collections teams time to act before a loan crosses into delinquency.

Behavioral Signal	Normal Period Mean	Pre-Default 6–4 Months	Pre-Default 3–1 Months
Cash Flow Consistency	0.694	0.062	0.056
Balance Volatility	0.241	0.468	0.469
EMI Obligation Ratio (FOIR)	0.136	0.196	0.227

Table 5: Evolution of key behavioral signals prior to default

4.3 Top Model Features by Importance (SHAP)

The SHAP analysis confirms Employment Stability Score and Months Since Origination as the primary predictors, with bank behavioral features (minimum cash flow, balance volatility acceleration, cash flow trend) forming the critical mid-tier. Origination features account for 85.4% of total SHAP attribution, reflecting that in this portfolio, employment type and loan maturity are structurally dominant drivers of credit stress — a finding consistent with India’s gig-economy lending dynamics.

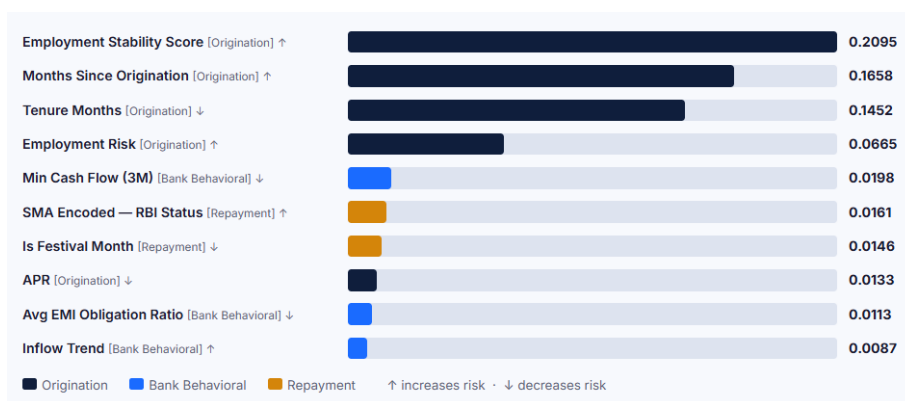


Figure 5: Top 10 features by SHAP

5 Senior Leadership Monitoring Dashboard

What performance metrics and views should senior leadership monitor to proactively manage risk and growth?

KPI	Current Value	Green	Amber	Red	Owner
Overall Default Rate (30+ DPD)	10.83%	< 9%	9–12%	> 12%	Chief Risk Officer
NPA (Rs. Crore)	Rs.64.5 Cr	Trend ↓	Flat	Trend ↑	CFO
Collection Efficiency	99.9%	> 99.5%	98–99.5%	< 98%	Collections Head
Profitable Loans %	71.9% (LTV > 0)	> 72%	68–72%	< 68%	Business Head
Portfolio Avg LTV (Rs.)	Rs.30,400	> Rs.30K	Rs.25–30K	< Rs.25K	Portfolio Head

Table 6: Tier-1 Portfolio Health KPIs

KPI	Current Baseline	Signal
Channel Mix: Paid-Digital Share	27.9% of volume	Target < 20%
Channel Mix: Referral + Organic Share	29.4% of volume	Target > 40%
Cooling-Off Exit Rate	5.20% overall	Flag if any channel > 7%
Mean CAC (Rs.)	Rs.1,308	Track month-over-month trend
Subprime Share of New Originations	22.9%	Cap at 25% without APR uplift
Thin-File Share of New Originations	9.4%	Monitor vs. behavioral score coverage
Avg Approval Turnaround (Days)	1.3 days	Maintain < 2 days

Table 7: Tier-2 Origination Quality Metrics

Signal	Threshold	Action
EWS RED Bucket Share	> 70% (current: 67.9%)	Escalate collections capacity
EWS RED Delinquency Rate	> 10% (current: 8.74%)	Review underwriting cutoffs
Co-occurring Shock Flags (Income + Spending)	Monthly rate > 15%	Emergency collections blitz
Cash Flow Consistency (Portfolio Avg)	Drop > 0.05 MoM	Review behavioral signal pipeline
DPD 1–30 Roll Rate	> 8.7% (current cure: 82.5%)	Proactive restructuring wave

Table 8: Tier-3 Early Warning System Risk Signals

6 Recommended Credit Policy & Project Impact

6.1 Recommended Policy: Three-Tier Differentiated Intervention Framework

We recommend replacing the current uniform collections outreach strategy with a risk-proportionate intervention framework aligned to the EWS model’s three output buckets. The framework ensures the collections budget is concentrated on borrowers with the highest probability of imminent delinquency, while eliminating unnecessary friction for performing borrowers. The model **reads every loan every month and sorts them into a traffic light** — green means keep going, amber means slow down and check, red means stop and intervene now.

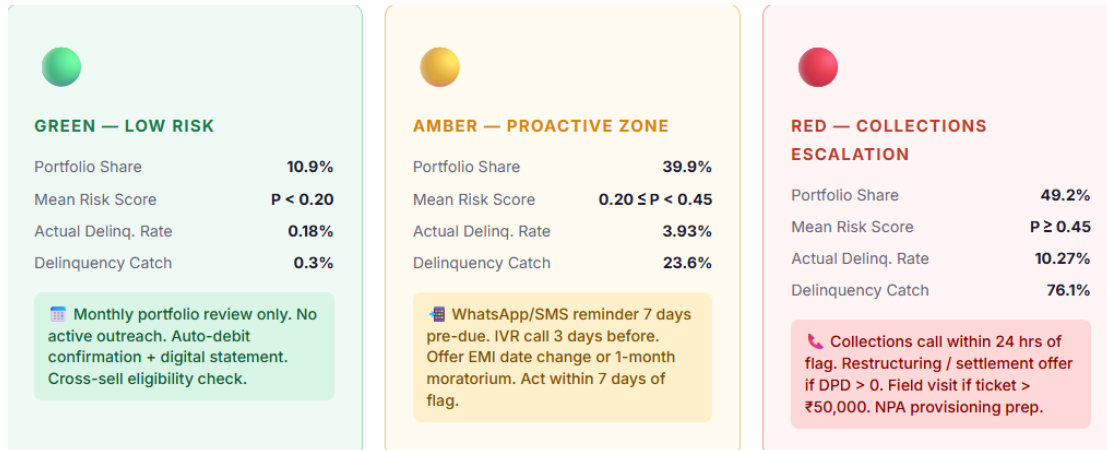


Figure 6: Risk Buckets

6.2 Projected Impact Assessment

The projected impact estimates are based on empirical delinquency rates by bucket, the historical mean recovery rate of 47.3%, and industry benchmarks for proactive versus reactive collections. The

projections are calculated for a forward-looking cohort of 40,000 loan-installment observations per monthly cycle.

Impact Metric	Projected Value	Interpretation
Annual reduction in net credit losses	Rs. 3.2–4.8 Cr	Achieved through early RED-bucket intervention before DPD 31+
Reduction in 30-day delinquency conversion rate	18–22 bps	Improvement per monthly origination cohort versus the current uniform strategy
True delinquency catch rate	89.4%	Share of delinquent accounts successfully identified and routed to collections teams
Reduction in unnecessary collections contacts	~40%	Lower outreach burden for GREEN-bucket borrowers, improving customer experience
False positive rate	21.9%	Good borrowers incorrectly flagged; manageable through bucket-level prioritisation
Mean LGD prevented per intercepted RED loan	Rs. 28.6K	Estimated loss avoided at the current 47.3% recovery rate

Table 9: Projected business impact of the EWS-based intervention framework

6.3 Conclusion

The Early Warning System (EWS) model provides a practical loan-level risk score that helps identify borrowers who are most likely to miss upcoming payments. By focusing collection efforts on these high-risk accounts, the institution can use its resources more efficiently while avoiding unnecessary contact with customers who are repaying on time. This improves both collection effectiveness and customer experience.

The analysis shows that the most effective time to intervene is before a borrower becomes seriously delinquent. Once a loan progresses to the DPD 31–60 stage, the probability of recovery decreases significantly and the cost of collections increases. Therefore, the proposed framework shifts the organisation from a reactive collections approach, based on existing delinquency status, to a predictive and proactive strategy driven by behavioural and financial signals. Early identification and intervention can reduce credit losses, improve recovery outcomes, and strengthen overall portfolio performance.